Course 0: generalities and evaluation

IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom

Discord server

- `https://discord.gg/ZuHcNMefaY` (use this link only to join the first time)
- Please immediately change your login to your first name and last name

Course organisation (1/5) – Schedule

Sessions

- **S1 (Sep 25):** Generalities + Lab 1 (Python, venv, numpy) + project definition
- **S2 (Oct 2):** Project validation presentations
- **S3 (Oct 9):** Supervised learning (course + lab)
- **S4 (Oct 16):** Unsupervised learning (course + lab)
- **S5 (Oct 23):** Deep learning + project milestone
- **Holiday break**
- **Exam (Nov 6):** courses 1–4
- **S6 (Nov 6):** Foundation models
- **S7 (Nov 13):** Adaptation of models
- **S8 (Nov 13):** Debate – societal impacts of AI
- **S9 (Nov 20, 27, Dec 4):** Project-dedicated sessions
- **S10 (Dec 11):** Final presentations

Typical session

- Short quiz on the previous lesson (10 min)
- Lesson (about 1h)
- Lab session
- Project work

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Course organisation (2/5) – Lab sessions

Lab sessions

- Python based
- scikit-learn for “classical” machine learning
- PyTorch for deep learning
- Hugging Face for foundation models

Each lab first uses a toy example, then you **apply the techniques to your own project data**.

You are expected to finish a Lab before the start of the next Lab session!

Course organisation (3/5) – Teams and project

Teams (binômes)

- Work in **binômes** (pairs)
- Split into **4 groups** of at most **7 teams**
- Trinômes / monômes accepted for odd counts (still max 7 teams per group)
- One **Discord channel per binôme**: pick a numbered channel and write both members' names

The project starts now!

- You choose your project in **Session 1**
- By default, define your own; a backup list is available on GitHub
- **Reserve your topic early**: post a one-line title in your channel, well before Session 2 (no two teams in a group on the same topic)

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Course organisation (4/5) – Evaluation

Evaluation moments

- **Wooclap quiz** at the start of each lesson (in the room, no documents, no AI)
- **Validation presentation** – Session 2
- **Exam** – Friday, November 6th, courses 1–4 (paper, two A4 manuscript pages, no electronic device)
- **Final presentation** – Session 10

Course organisation (5/5) – Presentations

Work in binômes.

Validation presentation (Session 2)

- **10 minutes**, questions included
- Present: task, data, metrics, and existing methods
- Attend your group's presentations and ask questions
- Not validated $\rightarrow$ re-take, with a penalty

Final presentation (Session 10)

- **15 minutes**, questions included
- Present your project definition, results, and your own analysis of the results
- Do not present everything: focus on the task definition and your most interesting results